

M002 Extension 1 — Redstone Trail Solver

Topic: Redstone Trail Solver (3D Cube Maze) · **Course:** Maze Madness · **Level:** Extension (Intermediate) · **Time:** about 45 minutes

The agent solves a **3D cube maze** on its own by reading a **redstone trail** all around it. Drive it with chat: **l** left, **r** right, **run** starts it, **rl** brings it back.

1 · Read the Chat Commands

A chat word binds to an action. Type the word, the agent runs the code.

```
def turn_left():
    agent.turn(LEFT_TURN)
player.on_chat("l", turn_left)
```

You type **l**. What does the agent do?


```
def go_back():
    agent.teleport_to_player()
player.on_chat("rl", go_back)
```

You type **rl**. Where does the agent go? When is that handy?

2 · Trace the Solver 🔍

This loop runs when you type `run`.

```
while True:
    agent.move(FORWARD, 1)
    if agent.detect(REDSTONE, DOWN):
        agent.move(UP, 1)
    elif agent.detect(REDSTONE, UP):
        agent.move(DOWN, 4)
```

`while True` keeps going. The maze is 3D. Which line moves the agent up a floor?

Which line moves it down? How far?

What does `detect(REDSTONE, DOWN)` tell the agent?

3 · Fill the Blanks

The agent should step forward, then go **up** when there is redstone **below** it.

```
agent.move(_____, 1)
if agent.detect(REDSTONE, _____):
    agent.move(_____, 1)
```

Word bank: FORWARD · DOWN · UP

4 · Spot the Bug 🐛

This should **keep going** and check the trail on every step.

```
agent.move(FORWARD, 1)
if agent.detect(REDSTONE, DOWN):
    agent.move(UP, 1)
```

Hint: it only checks once. What wraps a block so it repeats?

Write the fix:

One line: why was it wrong?

5 · Show Your Work 📷 🎥

Open the world and type `run`. Let the agent reach the goal.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.